

Statistical baseline model

Task-1 (DS-1)

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APMS 2014 Mental Health Survey

1. Data Description:

- Loaded 9 APMS 2014 Chapter files:
- CMD, PTSD, Psychotic Disorder, Personality Disorder, ADHD, Bipolar, Alcohol Use, Drug Dependence, Suicidal Behaviour.

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All 9 chapter files found in: /content/  
ch-02: Common Mental Disorders (CMD)  
ch-04: PTSD  
ch-05: Psychotic Disorder  
ch-07: Personality Disorder  
ch-08: ADHD  
ch-09: Bipolar Disorder  
ch-10: Alcohol Use  
ch-11: Drug Dependence  
ch-12: Suicidal Behaviour
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- Each chapter contained **multiple sheets** (e.g., 2.1, 4.1, etc.)

Chapter	Disorder	Sheets
ch-02	Common Mental Disorders (CMD)	['2.1', '2.2', '2.3', '2.4', '2.5', '2.6', '2.7', '2.8', '2.9', '2.10', '2.11', '2.12']
ch-04	PTSD	['4.1', '4.2', '4.3', '4.4', '4.5', '4.6', '4.7', '4.8', '4.9', '4.10', '4.11', '4.12', '4.13']
ch-05	Psychotic Disorder	['5.1', '5.2', '5.3', '5.4', '5.5', '5.6', '5.7', '5.8', '5.9', '5.10', '5.11', '5.12']
ch-07	Personality Disorder	['7.1', '7.2', '7.3', '7.4', '7.5', '7.6', '7.7', '7.8', '7.9', '7.10', '7.11', '7.12', '7.13', '7.14', '7.15', '7.16', '7.17', '7.18', '7.19', '7.20']
ch-08	ADHD	['8.1', '8.2', '8.3', '8.4', '8.5', '8.6', '8.7', '8.8', '8.9', '8.10', '8.11', '8.12']
ch-09	Bipolar Disorder	['9.1', '9.2', '9.3', '9.4', '9.5', '9.6', '9.7', '9.8', '9.9', '9.10']
ch-10	Alcohol Use	['10.1', '10.2', '10.3', '10.4', '10.5', '10.6', '10.7', '10.8', '10.9', '10.10', '10.11', '10.12', '10.13', '10.14', '10.15']
ch-11	Drug Dependence	['11.1', '11.2', '11.3', '11.4', '11.5', '11.6', '11.7', '11.8', '11.9', '11.10', '11.11', '11.12', '11.13', '11.14', '11.15', '11.16', '11.17', '11.18', '11.19', '11.20']
ch-12	Suicidal Behaviour	['12.1', '12.2', '12.3', '12.4', '12.5', '12.6', '12.7', '12.8', '12.9', '12.10', '12.11', '12.12', '12.13', '12.14']

2. Confidence Interval Formula:

- Since our data is in percentage format (e.g. "16.2%"), each value represents a **proportion** from a survey.
- The standard way to compute confidence intervals for proportions is the **Wilson Score method**.

Wilson Score CI – verification test

Disorder : Any CMD (North East, All Adults)

Prevalence: 16.2% Sample n: 395

95% CI : [12.8949% , 20.1562%]

CI Width : 7.2613 percentage points

SE : $\pm 1.8524\%$

Interpretation: We are 95% confident the true prevalence of Any CMD in North East lies between 12.8949% and 20.1562%.

3. Extracting Data from Each Chapter:

Each chapter's regional table has the same structure. We extract:

- Observed prevalence (%) for Men / Women / All Adults.
- Sample base sizes (n) for each region.
- Then compute Wilson Score CIs for each cell.

CMD data extracted: 189 rows

ANY CMD – All Adults – Observed Prevalence with 95% Confidence Intervals

	Region	Prevalence_Observed %	Sample_n	CI_Lower_95 %	CI_Upper_95 %	CI_width_pp
Yorkshire & the Humber	North East	16.2000	395.0000	12.8949	20.1562	7.2613
	North West	19.0000	1011.0000	16.7009	21.5338	4.8329
	Yorkshire & the Humber	16.2000	697.0000	13.6511	19.1194	5.4683
	East Midlands	16.7000	723.0000	14.1588	19.5932	5.4344
	West Midlands	18.6000	867.0000	16.1505	21.3265	5.1760
	East of England	14.1000	852.0000	11.9240	16.5983	4.6743
	London	18.9000	864.0000	16.4293	21.6461	5.2168
	South East	13.6000	1289.0000	11.8365	15.5798	3.7433
	South West	19.9000	848.0000	17.3512	22.7203	5.3691

PTSD data extracted: 54 rows

PTSD SCREEN POSITIVE – All Adults – 95% Confidence Intervals

	Region	Prevalence_Observed %	Sample_n	CI_Lower_95 %	CI_Upper_95 %	CI_width_pp
Yorkshire & the Humber	North East	3.6862	350.0000	2.1624	6.2157	4.0533
	North West	4.6727	938.0000	3.4971	6.2181	2.7210
	Yorkshire & the Humber	4.8754	600.0000	3.4210	6.9039	3.4829
	East Midlands	4.7779	662.0000	3.3978	6.6797	3.2819
	West Midlands	4.6485	772.0000	3.3747	6.3714	2.9967
	East of England	5.3255	808.0000	3.9779	7.0958	3.1179
	London	4.4861	754.0000	3.2251	6.2085	2.9834
	South East	3.2962	1185.0000	2.4210	4.4731	2.0521
	South West	4.0151	775.0000	2.8447	5.6392	2.7945

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Psychosis rows : 27
PD rows       : 27
ADHD rows     : 54
Bipolar rows  : 27
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Alcohol rows : 162
Drug rows    : 81
Suicide rows : 81
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4. Combining all data into Master Data Frame:

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MASTER DATASET SUMMARY
=====
Total rows      : 702
Disorder groups: 9
Disorders       : 26
Regions         : 9
Sex categories  : ['Men' 'Women' 'All adults']

Rows per Disorder Group:
Disorder_Group
ADHD                54
Alcohol Use         162
Bipolar Disorder    27
Common Mental Disorders 189
Drug Dependence     81
PTSD                54
Personality Disorder 27
Psychotic Disorder  27
Suicidal Behaviour  81
```

Sample of Master Data Frame:

Sample of master_df:

Chapter	Disorder_Group	Disorder	Sex	Region	Prevalence_Observed %	Prevalence_AgeStd %	Sample_n	CI_Lower_95 %	CI_Upper_95 %
02 Common Mental Disorders	Generalised anxiety disorder	Men	North East	5.7000	5.3000	155.0000	3.0101	10.5326	
02 Common Mental Disorders	Generalised anxiety disorder	Men	North West	6.4000	6.4000	422.0000	4.4357	9.1510	
02 Common Mental Disorders	Generalised anxiety disorder	Men	Yorkshire & the Humber	4.5000	4.7000	273.0000	2.6091	7.6536	
02 Common Mental Disorders	Generalised anxiety disorder	Men	East Midlands	4.3000	4.5000	318.0000	2.5631	7.1279	
02 Common Mental Disorders	Generalised anxiety disorder	Men	West Midlands	4.4000	4.5000	343.0000	2.6883	7.1218	
02 Common Mental Disorders	Generalised anxiety disorder	Men	East of England	5.5000	5.8000	348.0000	3.5547	8.4171	
02 Common Mental Disorders	Generalised anxiety disorder	Men	London	5.5000	5.9000	332.0000	3.5182	8.4998	
02 Common Mental Disorders	Generalised anxiety disorder	Men	South East	4.3000	4.1000	549.0000	2.8970	6.3381	
02 Common Mental Disorders	Generalised anxiety disorder	Men	South West	3.7000	3.5000	318.0000	2.1176	6.3876	
02 Common Mental Disorders	Depressive episode	Men	North East	1.5000	1.6000	155.0000	0.4483	4.8976	

5. Descriptive Statistics:

For each disorder:

- Mean
- Median
- Standard deviation
- Min / Max
- Range
- Coefficient of Variation (CV%)

Disorder_Group	Disorder	N_regions	Mean	Median	Std_Dev	Min	Max	Range	CV_pct
ADHD	ADHD screen (4+ symptoms)	9	9.6270	9.9310	1.3360	7.5210	11.6850	4.1640	13.8750
ADHD	ADHD screen (all 6 symptoms)	9	0.6500	0.6450	0.3130	0.3580	1.4030	1.0450	48.1360
Alcohol Use	Harmful/dependent drinking (AUDIT 8+)	9	19.7620	18.7650	3.1210	15.5310	25.4970	9.9660	15.7910
Alcohol Use	Harmful/mild dependence (AUDIT 16-19)	9	1.8570	1.7800	0.5640	0.6910	2.5750	1.8840	30.3810
Alcohol Use	Hazardous drinking (AUDIT 8-15)	9	16.6330	15.1900	3.0470	12.7390	22.3730	9.6330	18.3210
Alcohol Use	High-risk drinking (AUDIT 16+)	9	3.1290	3.1240	0.6160	2.1280	4.1170	1.9880	19.6840
Alcohol Use	Non-drinker/low risk (AUDIT 0-7)	9	80.2380	81.2350	3.1210	74.5030	84.4690	9.9660	3.8890
Alcohol Use	Probable dependence (AUDIT 20+)	9	1.2720	1.1410	0.3660	0.9760	2.0030	1.0270	28.7510
Bipolar Disorder	Bipolar disorder screen positive	9	1.9780	2.0820	0.6520	0.9770	2.7330	1.7570	32.9640
Common Mental Disorders	All phobias	9	2.3780	2.6000	0.6740	1.3000	3.1000	1.8000	28.3510
Common Mental Disorders	Any CMD	9	17.0220	16.7000	2.2340	13.6000	19.9000	6.3000	13.1220
Common Mental Disorders	CMD - NOS	9	7.7220	8.1000	1.6350	5.6000	9.5000	3.9000	21.1770
Common Mental Disorders	Depressive episode	9	3.3330	3.3000	0.7430	2.3000	4.6000	2.3000	22.2990
Common Mental Disorders	Generalised anxiety disorder	9	5.9670	5.8000	0.8190	4.9000	7.2000	2.3000	13.7180
Common Mental Disorders	Obsessive compulsive disorder	9	1.3330	1.1000	0.6820	0.6000	2.7000	2.1000	51.1430
Common Mental Disorders	Panic disorder	9	0.5560	0.4000	0.4160	0.1000	1.4000	1.3000	74.8200
Drug Dependence	Any drug dependence	9	2.8240	2.6000	0.8570	1.8810	4.6600	2.7790	30.3410
Drug Dependence	Cannabis only dependence	9	2.0630	1.9810	0.6130	1.2180	3.3070	2.0890	29.7150
Drug Dependence	Other drug(s) dependence	9	0.7610	0.6190	0.3820	0.3870	1.3660	0.9790	50.2270
PTSD	PTSD screen positive	9	4.4200	4.6480	0.6370	3.2960	5.3250	2.0290	14.3990
PTSD	Trauma experienced	9	31.4210	31.7040	3.0850	25.1120	34.7310	9.6190	9.8170
Personality Disorder	Any personality disorder (SAPAS)	9	13.7770	13.9450	1.3800	11.1450	16.1780	5.0340	10.0170
Psychotic Disorder	Psychotic disorder (past year)	9	0.5370	0.5780	0.1690	0.1990	0.8490	0.6490	31.4690
Suicidal Behaviour	Lifetime self-harm	9	7.3390	7.1090	0.8790	6.0180	8.5640	2.5460	11.9830
Suicidal Behaviour	Lifetime suicidal thoughts	9	20.5390	20.9200	1.8110	17.6320	23.4990	5.8670	8.8160
Suicidal Behaviour	Lifetime suicide attempts	9	6.8900	7.0150	0.7160	5.8550	7.9800	2.1250	10.3860

6. Region-wise Statistical Tables:

Generated separate tables for:

- CMD
- PTSD
- Psychotic Disorder
- Personality Disorder
- Bipolar Disorder
- Drug Dependence

Each Table include:

- Region
- Prevalence %
- Sample size (n)
- CI lower & upper
- CI width
- SE%

ANY CMD -- Common Mental Disorders						
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width	SE%
North East	16.200%	395	12.895%	20.156%	7.261pp	1.852%
North West	19.000%	1011	16.701%	21.534%	4.833pp	1.233%
Yorkshire & the Humber	16.200%	697	13.651%	19.119%	5.468pp	1.395%
East Midlands	16.700%	723	14.159%	19.593%	5.434pp	1.386%
West Midlands	18.600%	867	16.151%	21.326%	5.176pp	1.321%
East of England	14.100%	852	11.924%	16.598%	4.674pp	1.192%
London	18.900%	864	16.429%	21.646%	5.217pp	1.331%
South East	13.600%	1289	11.836%	15.580%	3.743pp	0.955% *
South West	19.900%	848	17.351%	22.720%	5.369pp	1.370% *
ENGLAND (weighted avg) (* = highest or lowest region)	16.999%	7546	16.168%	17.863%	1.695pp	0.432%

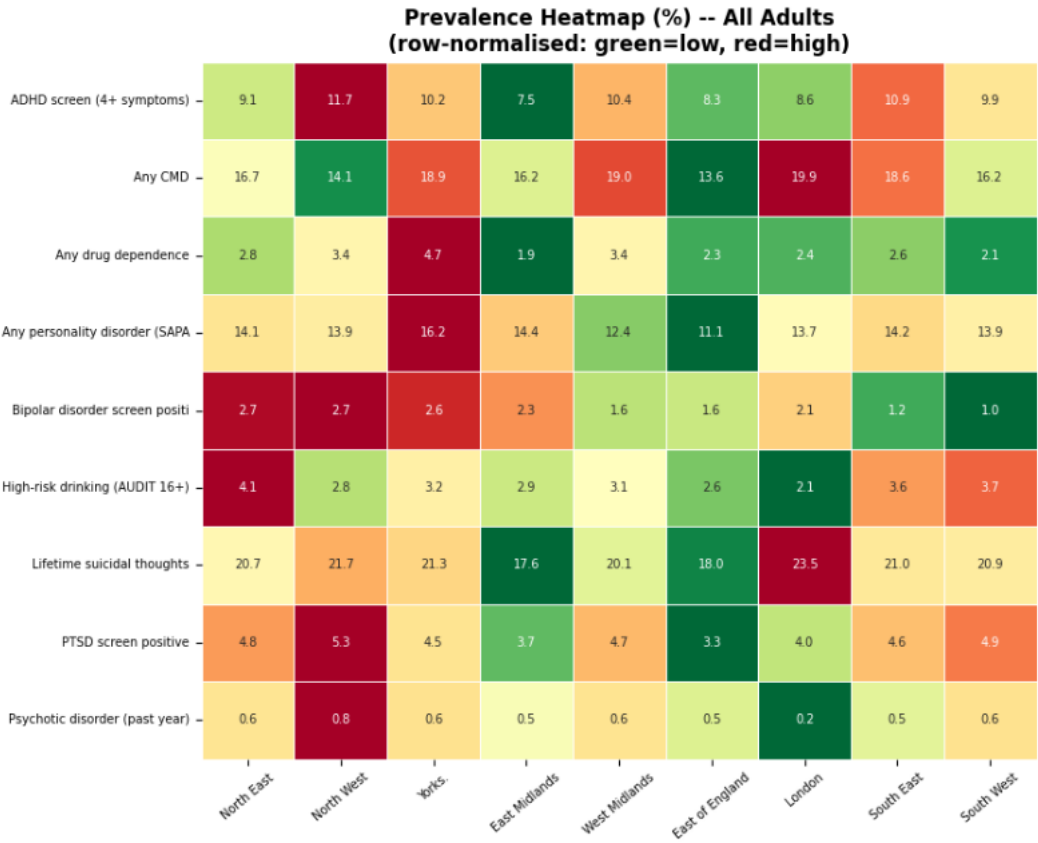
PTSD SCREEN POSITIVE -- PTSD						
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width	SE%
North East	3.686%	350	2.162%	6.216%	4.053pp	1.034%
North West	4.673%	938	3.497%	6.218%	2.721pp	0.694%
Yorkshire & the Humber	4.875%	599	3.421%	6.904%	3.483pp	0.888%
East Midlands	4.778%	661	3.398%	6.680%	3.282pp	0.837%
West Midlands	4.648%	771	3.375%	6.371%	2.997pp	0.764%
East of England	5.325%	808	3.978%	7.096%	3.118pp	0.795% *
London	4.486%	754	3.225%	6.208%	2.983pp	0.761%
South East	3.296%	1184	2.421%	4.473%	2.052pp	0.523% *
South West	4.015%	774	2.845%	5.639%	2.795pp	0.713% *
ENGLAND (weighted avg) (* = highest or lowest region)	4.403%	6843	3.942%	4.915%	0.973pp	0.248%

PSYCHOTIC DISORDER (PAST YEAR) -- Psychotic Disorder						
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width	SE%
North East	0.513%	810	0.203%	1.290%	1.087pp	0.277%
North West	0.580%	2067	0.332%	1.011%	0.679pp	0.173%
Yorkshire & the Humber	0.584%	1455	0.302%	1.126%	0.824pp	0.210%
East Midlands	0.578%	1367	0.292%	1.140%	0.849pp	0.216%
West Midlands	0.478%	1610	0.239%	0.952%	0.713pp	0.182%
East of England	0.849%	1666	0.507%	1.416%	0.908pp	0.232% *
London	0.585%	1617	0.313%	1.091%	0.778pp	0.199%
South East	0.466%	2428	0.262%	0.826%	0.563pp	0.144%
South West	0.199%	1555	0.069%	0.575%	0.506pp	0.129% *
ENGLAND (weighted avg) (* = highest or lowest region)	0.539%	14579	0.433%	0.672%	0.239pp	0.061%

ANY PERSONALITY DISORDER (SAPAS) -- Personality Disorder						
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width	SE%
North East	14.371%	360	11.123%	18.372%	7.248pp	1.849%
North West	12.415%	967	10.485%	14.644%	4.159pp	1.061%
Yorkshire & the Humber	13.904%	621	11.404%	16.848%	5.444pp	1.389%
East Midlands	14.084%	697	11.698%	16.864%	5.166pp	1.318%
West Midlands	14.217%	802	11.970%	16.805%	4.834pp	1.233%
East of England	13.945%	826	11.749%	16.474%	4.726pp	1.206%
London	16.178%	783	13.765%	18.922%	5.157pp	1.315% *
South East	11.145%	1224	9.502%	13.031%	3.529pp	0.900% *
South West	13.730%	802	11.520%	16.285%	4.765pp	1.216%
ENGLAND (weighted avg) (* = highest or lowest region)	13.667%	7082	12.887%	14.487%	1.600pp	0.408%

BIPOLAR DISORDER SCREEN POSITIVE -- Bipolar Disorder						
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width	SE%
North East	2.287%	360	1.174%	4.407%	3.233pp	0.825%
North West	1.578%	967	0.962%	2.576%	1.614pp	0.412%
Yorkshire & the Humber	0.977%	621	0.450%	2.106%	1.656pp	0.422% *
East Midlands	2.695%	698	1.728%	4.180%	2.451pp	0.625%
West Midlands	1.239%	802	0.673%	2.270%	1.597pp	0.407%
East of England	2.733%	825	1.821%	4.084%	2.263pp	0.577% *
London	2.591%	780	1.688%	3.960%	2.272pp	0.580%
South East	1.623%	1221	1.051%	2.499%	1.448pp	0.369%
South West	2.082%	802	1.298%	3.322%	2.024pp	0.516%
ENGLAND (weighted avg)	1.957%	7076	1.659%	2.307%	0.647pp	0.165%
(* = highest or lowest region)						

ANY DRUG DEPENDENCE -- Drug Dependence						
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width	SE%
North East	1.881%	348	0.893%	3.916%	3.022pp	0.771% *
North West	3.366%	954	2.396%	4.710%	2.315pp	0.591%
Yorkshire & the Humber	2.139%	598	1.249%	3.640%	2.391pp	0.610%
East Midlands	2.754%	691	1.771%	4.260%	2.489pp	0.635%
West Midlands	2.600%	790	1.699%	3.960%	2.261pp	0.577%
East of England	3.365%	798	2.321%	4.855%	2.534pp	0.646%
London	4.660%	760	3.377%	6.399%	3.022pp	0.771% *
South East	2.294%	1197	1.586%	3.306%	1.720pp	0.439%
South West	2.356%	785	1.504%	3.672%	2.167pp	0.553%
ENGLAND (weighted avg)	2.962%	6925	2.588%	3.388%	0.800pp	0.204%
(* = highest or lowest region)						



7. Composite Mental Health Burden Score:

No single disorder gives us full idea. To rank regions by overall mental health burden, So:

- Take the observed prevalence for each disorder in each region (All Adults only).
- Standardise each disorder to a **z-score**: $z = (\text{region_value} - \text{national_mean}) / \text{national_std}$.
- Average the z-scores across all disorders for each region.
- A positive score means above-average burden; negative means below average.

COMPOSITE MENTAL HEALTH BURDEN SCORE BY REGION (Higher score = more disorders above England average)						
Rank	Region	Z-Score		Category	Population	
1	London	+	0.7826	High	6,806,412	>>>>>>
2	East of England	+	0.6133	High	4,875,281	>>>>>
3	East Midlands	+	0.3553	High	3,782,353	>>>
4	West Midlands	+	0.1726	Above average	4,599,064	>
5	North West	+	0.0994	Above average	5,790,512	
6	Yorkshire & the Humber	-	0.0198	Below average	4,347,165	
7	South West	-	0.2792	Below average	4,473,319	>>
8	North East	-	0.6064	Low	2,154,854	>>>>>
9	South East	-	1.1178	Low	7,184,102	>>>>>>>>

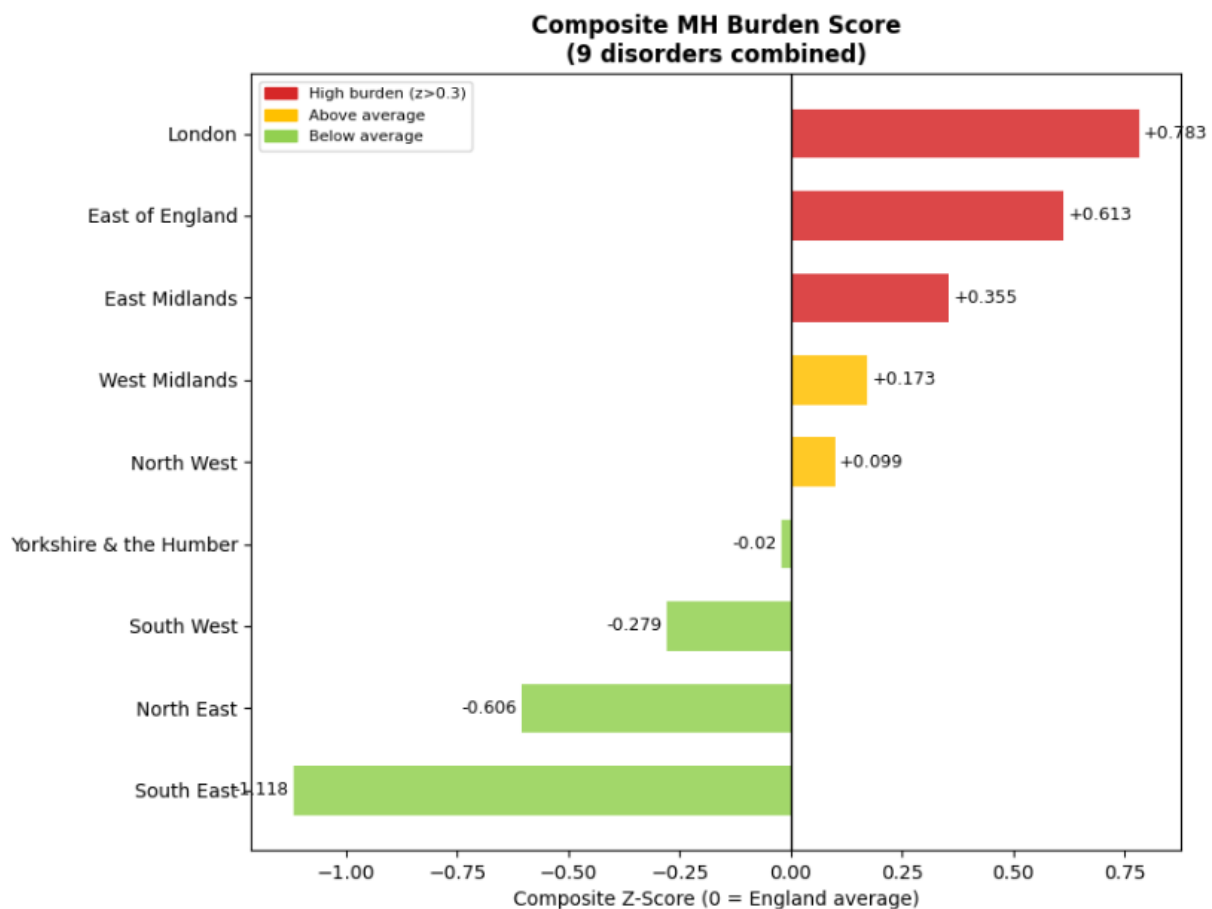
Z-score: 0 = England average. +1 = 1 std dev above average.

Interpretations:

- 0 → England average.
- Positive → above average burden.
- Negative → below average.

Ranking:

- London → Highest burden
- East of England
- East Midlands
- West Midlands
- North West
- Yorkshire & Humber
- South West
- North East
- South East → Lowest



8. England Summary:

ENGLAND NATIONAL SUMMARY – Weighted Prevalence Estimates with 95% CI

Disorder_Group	Disorder	England_Avg_%	Total_n	CI_Lower_%	CI_Upper_%	CI_width_pp
Common Mental Disorders	Any CMD	16.9990	7546	16.1681	17.8630	1.6950
	PTSD	4.4030	6843	3.9418	4.9151	0.9730
Psychotic Disorder	Psychotic disorder (past year)	0.5390	14579	0.4328	0.6720	0.2390
Personality Disorder	Any personality disorder (SAPAS)	13.6670	7082	12.8870	14.4870	1.6000
ADHD	ADHD screen (4+ symptoms)	9.7410	7542	9.0920	10.4307	1.3390
Bipolar Disorder	Bipolar disorder screen positive	1.9570	7076	1.6595	2.3069	0.6470
Alcohol Use	High-risk drinking (AUDIT 16+)	3.0950	7546	2.7276	3.5103	0.7830
Drug Dependence	Any drug dependence	2.9620	6925	2.5878	3.3879	0.8000
Suicidal Behaviour	Lifetime suicidal thoughts	20.5690	7544	19.6721	21.4962	1.8240

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	Lowest_Region	Highest_Region	Range_pp
Yorkshire & the Humber	South East (13.60%)	South West (19.90%)	6.3000
	South East (3.30%)	East of England (5.33%)	2.0290
	South West (0.20%)	East of England (0.85%)	0.6490
	South East (11.14%)	London (16.18%)	5.0340
	North East (7.52%)	East of England (11.69%)	4.1640
	South West (2.13%)	East Midlands (4.12%)	1.9880
	North East (1.88%)	London (4.66%)	2.7790
	North East (17.63%)	South West (23.50%)	5.8670

9. Final Conclusions:

- Mental health disorders are **highly prevalent**, with ~17% of adults affected by CMD (≈ 1 in 6 people).
- **Significant regional disparities** exist in mental health burden across England.
- **London and East of England** show the highest overall mental health burden.
- **South East and North East** show comparatively lower burden levels.
- **CMD (anxiety, depression)** is the most dominant contributor to overall mental health issues.
- **Suicidal thoughts (~20%+)** are consistently high across all regions, indicating a national concern.
- **Substance use disorders** (alcohol, drugs) are more prominent in urban regions.
- Low-prevalence disorders (psychotic, bipolar) are **less common but clinically critical**.
- Confidence intervals confirm that results for common disorders are **statistically reliable**.
- Composite Z-score analysis validates consistent patterns of regional mental health burden.
- Overall, mental health in England is **widespread, unevenly distributed, and requires targeted intervention**.
- Final Report: [APMS2014 MH Regional Statistical Model Report](#)